

Computer Networks

Lecture 16: Data Link Layer

Content

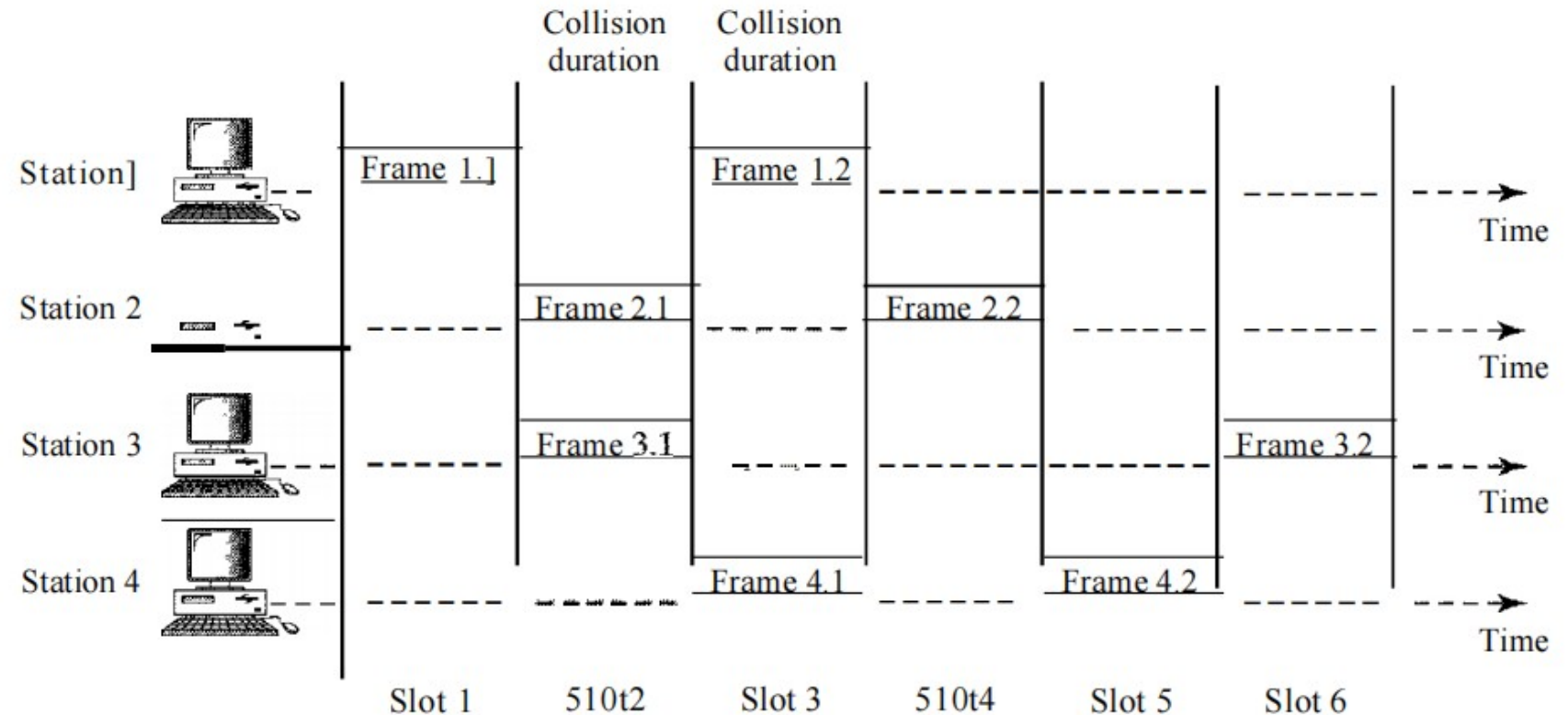


- **Slotted ALOHA**
- **CSMA**

Slotted ALOHA

Frame Transmission

In slotted ALOHA we divide the time into slots of T_{fr} s and force the station to send only at the beginning of the time slot.

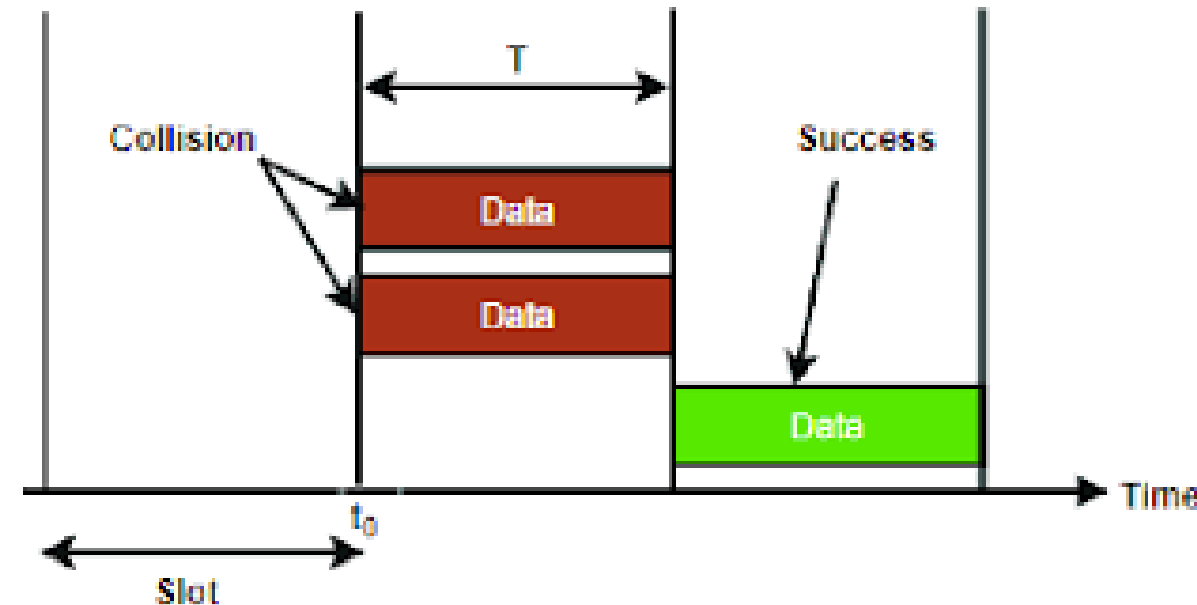


Slotted ALOHA

Vulnerable time

In a slotted system, transmission is allowed only at the beginning of synchronized time slots, which reduces collision probability.

- Collision occurs only if two stations transmit at the same slot start.
- The vulnerable time is reduced to half compared to unslotted transmission.
- Therefore, the total vulnerable period is T .



Slotted ALOHA

Throughput

In Slotted ALOHA, throughput represents the fraction of successful transmissions per time slot.

It can be proved that the average number of successful transmissions for slotted ALOHA is

$$S = G \times e^{-G}$$

The maximum throughput $S_{max} = 0.368$ when $G = 1$.

CSMA

In shared communication networks, multiple stations attempt to use the same transmission medium. When two or more stations transmit simultaneously, collisions occur, leading to data loss and reduced performance.

To address this issue, Carrier Sense Multiple Access (CSMA) was developed.

CSMA is a medium access control protocol in which a station:

- Listens to the channel before transmitting
- Transmits only if the medium is free
- Waits if the channel is busy

CSMA

CSMA requires a station to listen to (sense) the channel before transmitting.

- If the channel is idle → Transmit
- If the channel is busy → Wait and try again

CSMA can reduce the possibility of collision, but it cannot eliminate it.

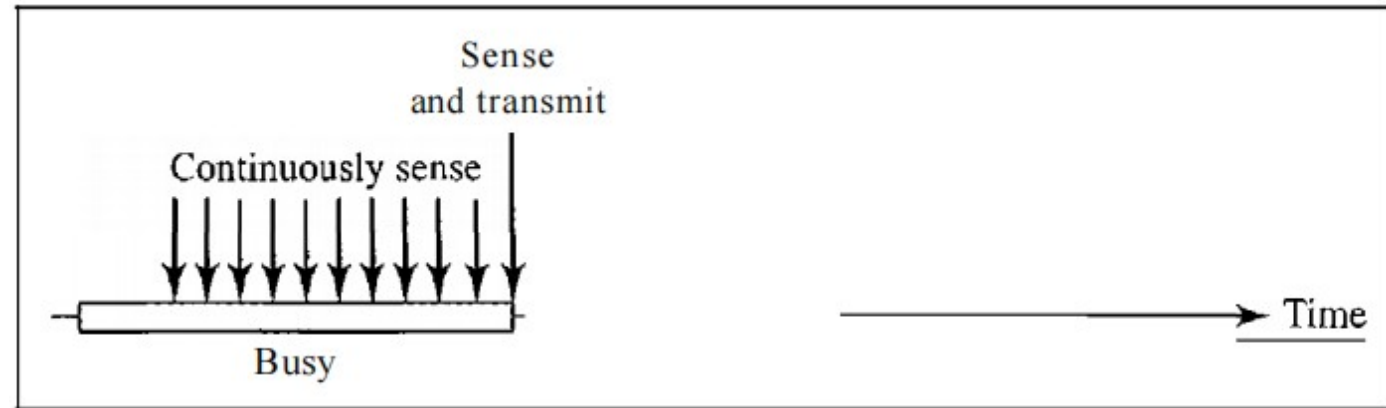


Persistence Method

What should a station do if the channel is busy? What should a station do if the channel is idle?

1-persistent method, In this a station first checks if the channel is free.

- If the channel is **idle (free)** → it **transmits immediately**.
- If the channel is **busy** → it keeps listening and waits until the channel becomes free.



The station sends the frame with **probability 1** (that means it will definitely send once the channel is free).

Persistence Method

What should a station do if the channel is busy? What should a station do if the channel is idle?

Non-persistent method, in this a station first checks if the channel is free.

- If the channel is idle → it transmits immediately.
- If the channel is busy → it does NOT keep listening continuously.
Instead, it waits for a random amount of time, then checks the channel again.

Because each station waits for a different random time, it is unlikely that two stations will transmit at the same time.

Sometimes the channel becomes free, but stations are still waiting during their random delay.

Persistence Method

What should a station do if the channel is busy? What should a station do if the channel is idle?

P-persistent method The p-persistent method is used in networks where time is divided into small time slots.

When a station finds the channel idle, it does not always send immediately.

Instead:

1. It sends the frame with probability p (for example, 0.5 or 0.3).
2. With probability $(1 - p)$, it waits for the next time slot and checks again.

At the next slot:

- If the channel is still idle, it again decides using probability p .
- If the channel is busy, it behaves as if a collision happened and waits using a backoff method.

CSMA/CD

In CSMA, we learned that stations sense the channel before transmitting to reduce collisions. However, collisions can still occur due to propagation delay.

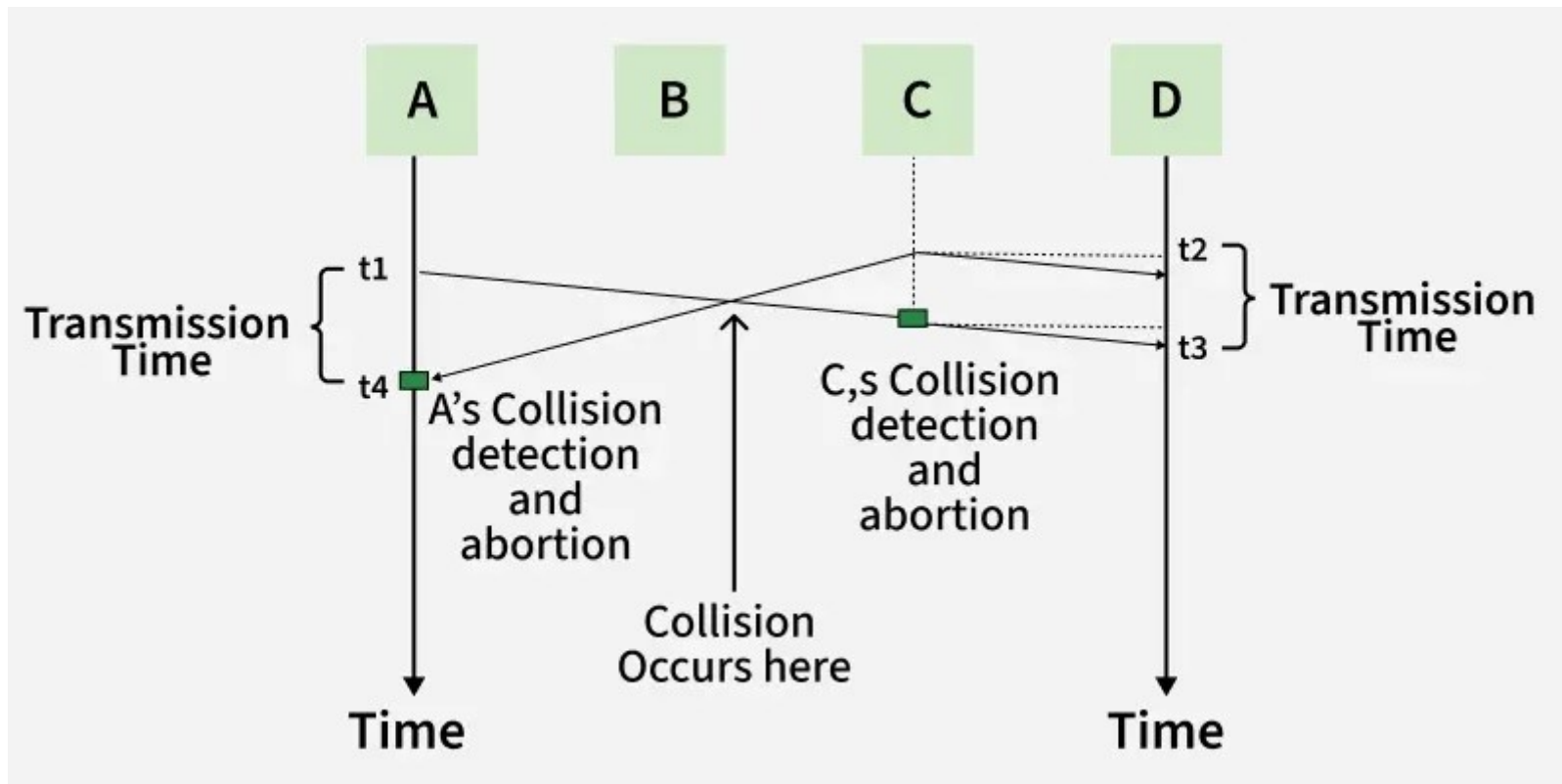
To further improve performance, Carrier Sense Multiple Access with Collision Detection (CSMA/CD) was introduced.

In CSMA/CD, a station:

- Listens before transmitting (same as CSMA)
- Continues listening while transmitting
- Detects a collision immediately if it occurs
- Stops transmission at once instead of sending the whole frame

CSMA/CD

Scenario



CSMA/CD

Minimum Frame Size

Consider two stations that are the maximum distance apart in the network:

- Station *A* starts transmitting.
- It takes T_p (propagation time) for the signal to reach Station *B*.
- If Station *B* also transmits, a collision occurs.
- The collision effect takes another T_p time to travel back to Station *A*.

So, total time for Station *A* to detect collision: $2T_p$

CSMA/CD

Minimum Frame Size

To ensure collision detection: $T_{fr} \geq 2T_p$

where:

T_{fr} = Frame transmission time

T_p = Maximum propagation time

- The station must still be transmitting after time $2T_p$.
- If transmission ends before $2T_p$, collision may go undetected.
- Therefore, the minimum frame transmission time must be at least twice the propagation delay.

Question

1. A network using CSMA/CD has a bandwidth of $10Mbps$. If the maximum propagation time is $25.6\mu S$, what is the minimum size of the frame?
2. If the distance between sender and receiver is $4KM$ velocity is $2 \times 10^8 m/s$, bandwidth is $10Mbps$. Find the minimum frame size that can be transmitted on CSMA/CD network

CSMA/CD

Energy Level In a wired CSMA/CD system (like traditional Ethernet), the cable can show three energy levels:

Zero Energy Level (Idle State)

- No station is transmitting.
- The cable has no signal.

Normal Energy Level (Single Transmission)

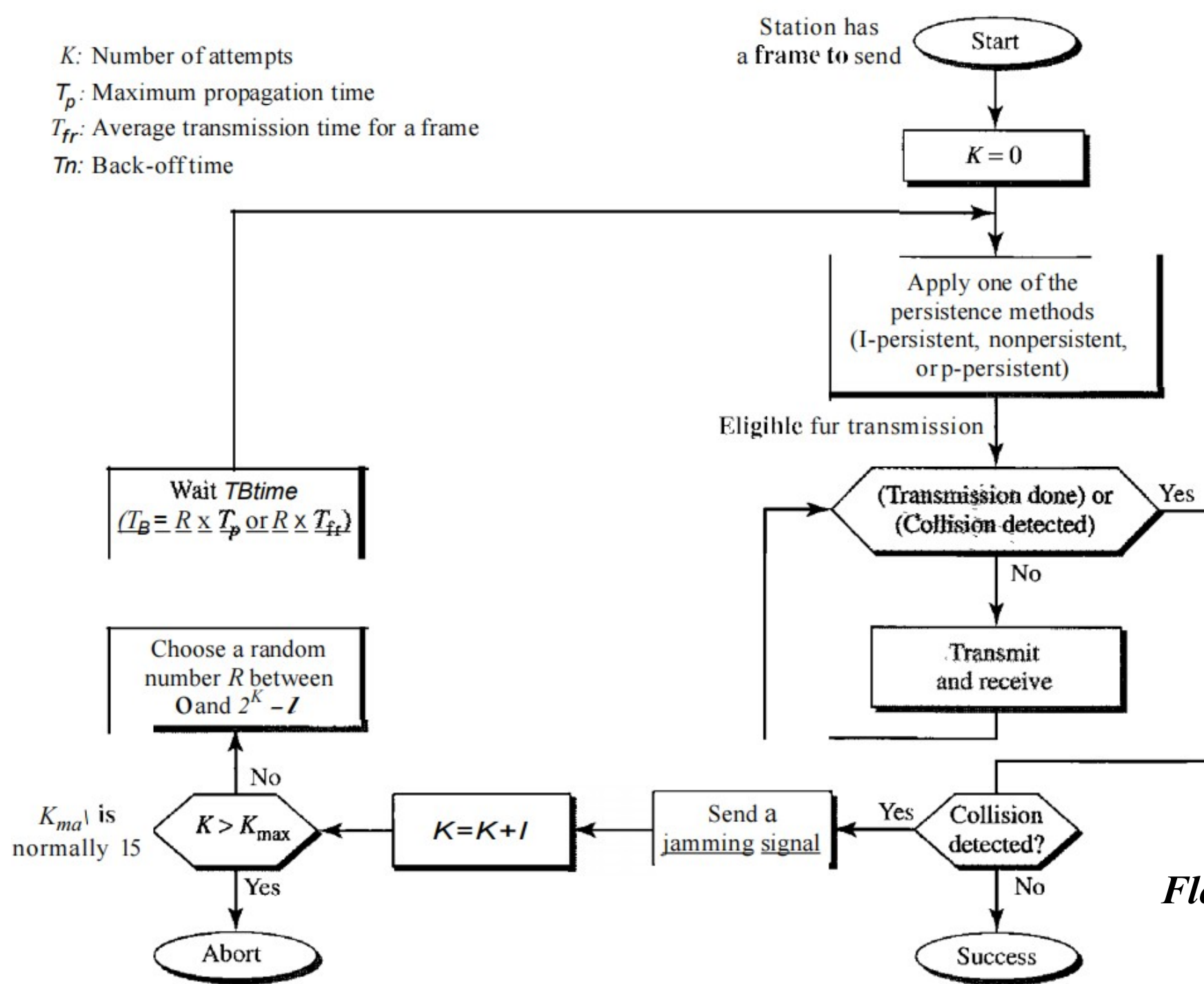
- Only one station is transmitting.
- The cable carries a normal signal voltage.

High Energy Level (Collision State)

- Two or more stations transmit at the same time.
- Signals overlap and add together

K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_n : Back-off time

CSMA/CD



Flow diagram for the CSMA/CD

CSMA/CD

The throughput of CSMA/CD is higher than pure ALOHA and slotted ALOHA.

Maximum throughput depends on the persistence method used

For 1-Persistent Method

- Maximum throughput $\approx 50\%$
- Occurs when $G = 1$

For Non-Persistent Method

- Maximum throughput can reach up to 90%
- Occurs when G is between 3 and 8

For P-Persistent Method

- Maximum throughput $\approx 70\%-80\%$
- Depends on P and G